Training session on Global Sensitivity Analysis
Part I. Variance-based methods

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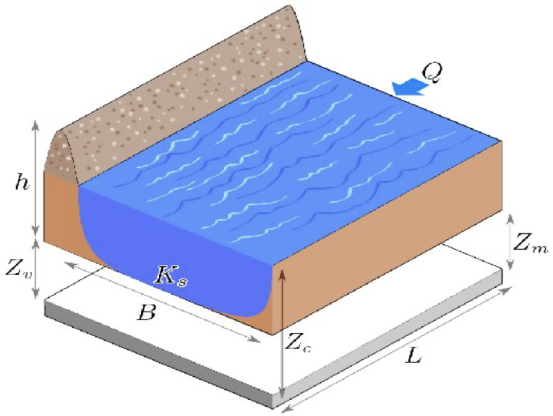
PILearnWater Chair, 2026 March 11

Part I

Introduction

A case study for global sensitivity analysis



A simplified flood model [Iooss, 2011], [Iooss and Lemaitre, 2015].

- Output 1: maximal annual overflow S (in meters), obtained from simplified hydro-dynamical equations of Saint-Venant:

$$S = Z_v + H - H_d - C_b \quad \text{with} \quad H = \left(\frac{Q}{BK_s \sqrt{\frac{Z_m - Z_v}{L}}} \right)^{0.6}$$

- Output 2: the cost (in million euros) of the damage on the dyke Y:

$$C = 1_{S>0} + \left[0.2 + 0.8 \left(1 - \exp^{-\frac{1000}{S^4}} \right) \right] 1_{S \leq 0} + \frac{1}{20} \left(H_d 1_{H_d > 8} + 81_{H_d \leq 8} \right)$$

A case study for global sensitivity analysis

- 8 inputs variables assumed to be independent r.v., with distributions:

Input	Description	Unit	Probability distribution
$X_1 = Q$	Maximal annual flowrate	m^3/s	Gumbel $\mathcal{G}(1013, 558)$, truncated on $[500, 3000]$
$X_2 = K_s$	Strickler coefficient	-	Normal $\mathcal{N}(30, 8^2)$, truncated on $[15, +\infty[$
$X_3 = Z_v$	River downstream level	m	Triangular $\mathcal{T}(49, 50, 51)$
$X_4 = Z_m$	River upstream level	m	Triangular $\mathcal{T}(54, 55, 56)$
$X_5 = H_d$	Dyke height	m	Uniform $\mathcal{U}[7, 9]$
$X_6 = C_b$	Bank level	m	Triangular $\mathcal{T}(55, 55.5, 56)$
$X_7 = L$	River stretch	m	Triangular $\mathcal{T}(4990, 5000, 5010)$
$X_8 = B$	River width	m	Triangular $\mathcal{T}(295, 300, 305)$

- *Aim: To detect unessential X_i's, to quantify the influence of X_i's on S, . . .*

A case study for global sensitivity analysis

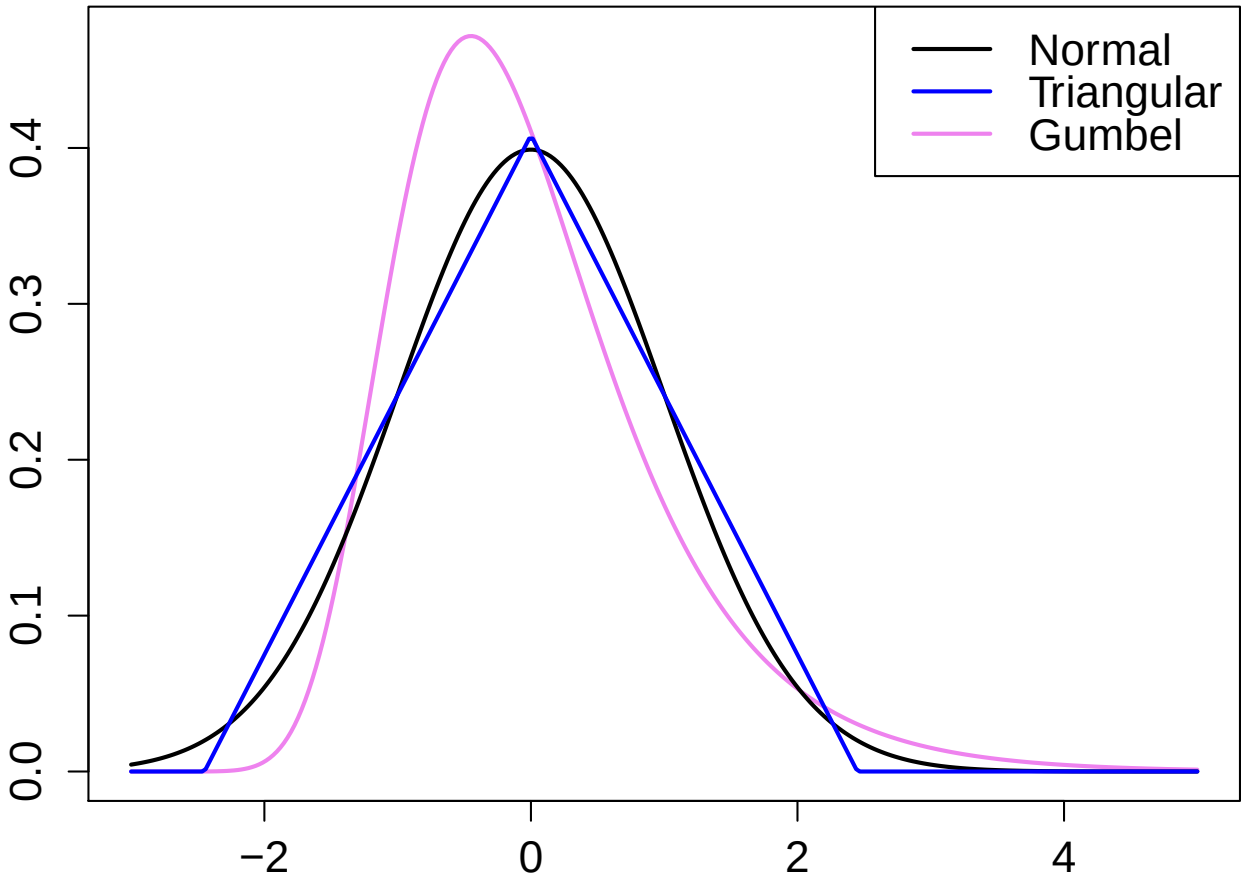
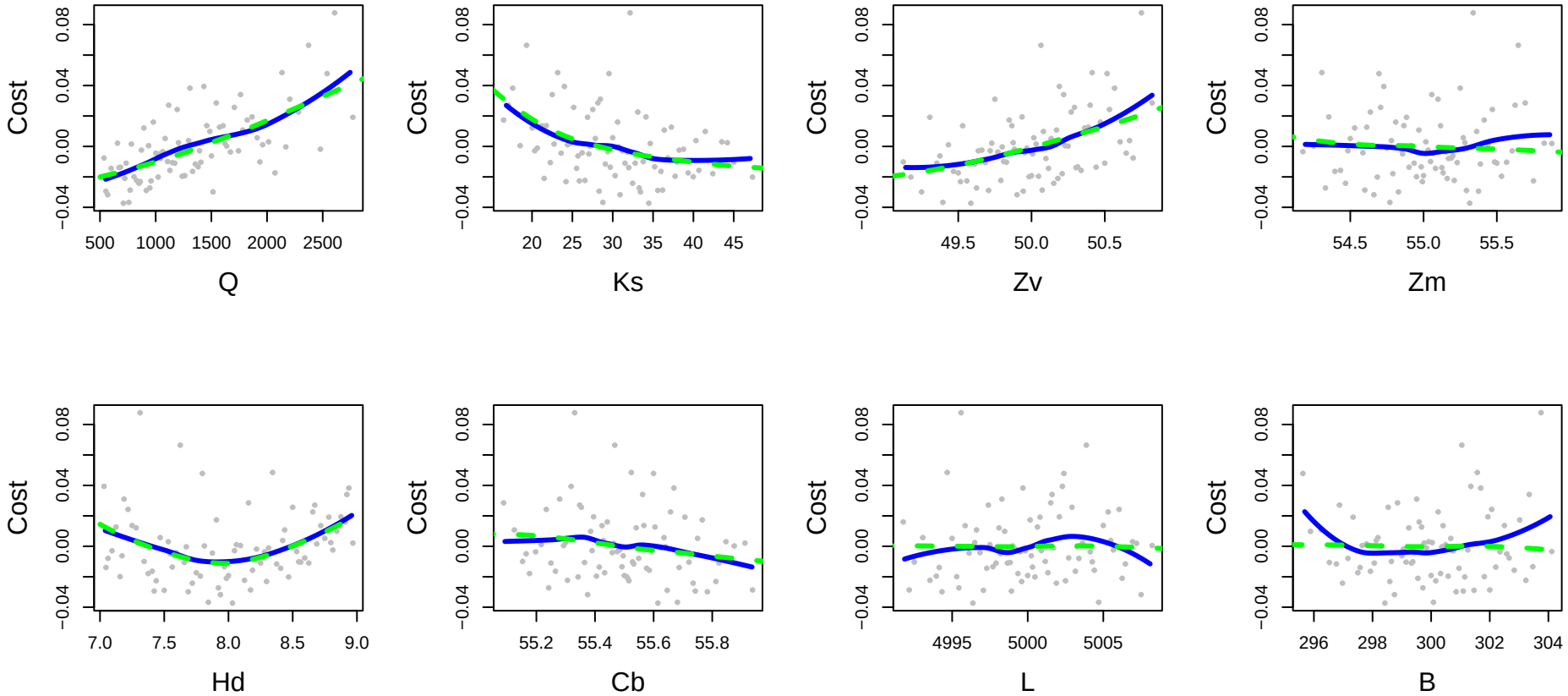


Figure: The 3 distributions types of the case study, here with mean 0 and variance 1

Qualitative study: Main effects



Plot of $f(X)$ versus X_i, and estimation of $x_i \mapsto \mathbb{E}(f(X)|X_i = x_i) - \mathbb{E}(f(X))$.

blue: directly with f and a large budget of $N = 10\,000$ runs.
green: with a metamodel (here Gaussian process) built with only $n = 80$ runs.

Quantitative study

Let $f : x = (x_1, \ldots, x_d) \mapsto f(x)$ some multivariate function.

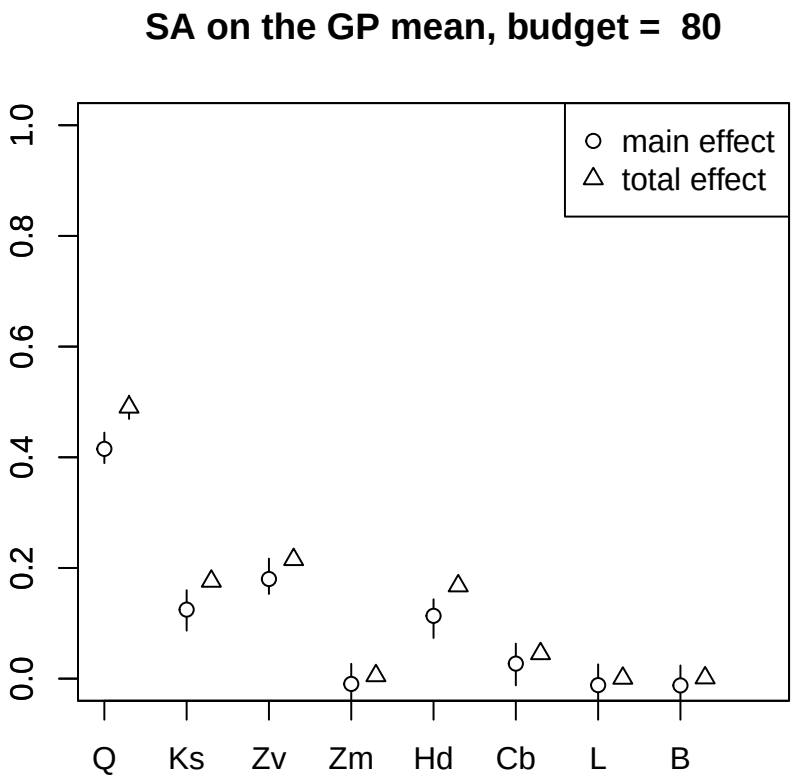
Questions

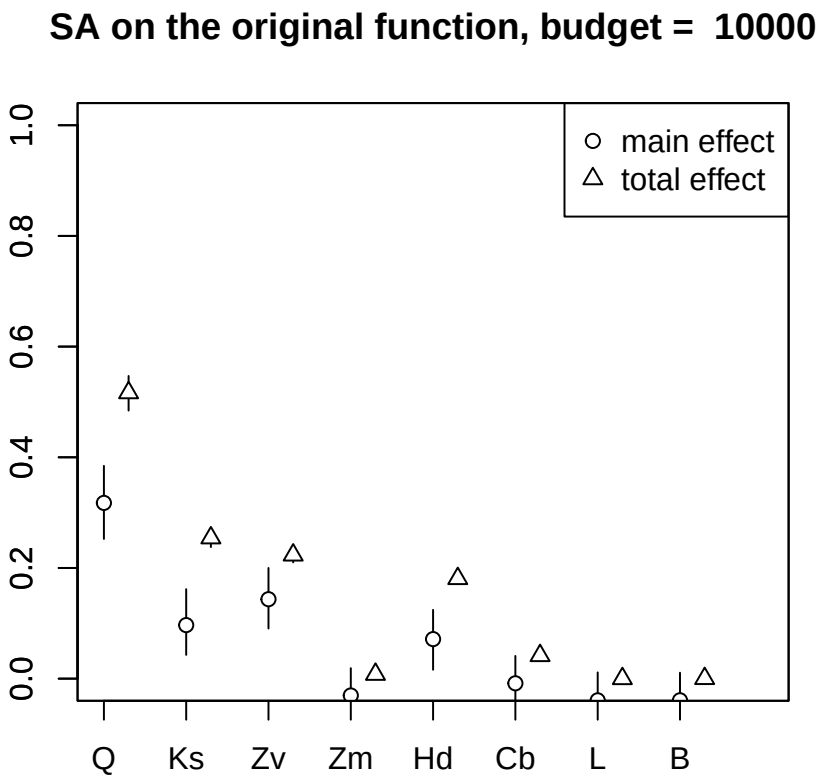
- *Screening*: what are the variables x_i that have no influence on $f(x)$?
- *Uncertainty quantification*: what is the influence of x_i on $f(x)$?

Answers

- Local and deterministic
  $\rightarrow$ Compute the *partial derivative at x_0*
- Global and probabilistic
  $\rightarrow$ Assess the influence of x_i, viewed as random var., on the *law* of $f(x)$
  $\rightarrow$ First step: look at the influence of x_i onto the *variance* of $f(x)$

Quantitative study





Estimated Sobol indices and total Sobol indices for the Cost function.
Sobol index : % of variance of $f(x)$ explained by x_i (alone)
Total Sobol index: % of variance of $f(x)$ explained by x_i, at least.

Outline of the training session

1. The rest of this talk: method based on *variance decomposition*
2. Next talk: construction of *chaos expansions metamodels* and why they are suitable for global sensitivity analysis
3. Last talk: *gradient-enhanced global sensitivity analysis*, or what we can do (still globally) when the gradient of f is available.

Two *reference books* to go further

- [Saltelli et al., 2008]
- [Da Veiga et al., 2021]

Part II

Variance-based global sensitivity analysis

Sobol-Hoeffding decomposition

Framework. $X = (X_1, \ldots, X_d)$ is a vector of independent input variables with distribution $\mu_1 \otimes \cdots \otimes \mu_d$, and $g : \Delta \subseteq \mathbb{R}^d \to \mathbb{R}$ is such that $g(X) \in L^2(\mu)$.

Theorem [Hoeffding, 1948, Efron and Stein, 1981, Sobol, 1993]

There exists a unique expansion of g of the form

$$g(X) = g_0 + \sum_{i=1}^{d} g_i(X_i) + \sum_{1 \leq i < j \leq d} g_{i,j}(X_i, X_j) + \cdots + g_{1,\ldots,d}(X_1, \ldots, X_d)$$

such that $E[g_I(X_I)|X_J] = 0$ for all $I \subseteq \{1, \ldots, d\}$ and all $J \subsetneq I$.

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such that $E[g_I(X_I)|X_J] = 0$ for all $I \subseteq \{1, \ldots, d\}$ and all $J \subsetneq I$. Moreover:

$$\begin{aligned}
g_0 &= \mathbb{E}[g(X)] \\
g_i(X_i) &= \mathbb{E}[g(X)|X_i] - g_0 \\
g_I(X_I) &= \mathbb{E}[g(X)|X_I] - \sum_{J \subsetneq I} g_J(X_J) \qquad \text{(recursion)} \\
&= \sum_{J \subseteq I} (-1)^{|I|-|J|} \mathbb{E}[g(X)|X_J] \qquad \text{(inclusion-exclusion)}
\end{aligned}$$

The non-overlapping condition

- The non-overlapping condition

$$\mathbb{E}[g_I(X_I)|X_J] = 0 \quad \text{for all} \quad J \subsetneq I$$

avoids one term to be considered as a more complex one.

- **Example.** Let $d = 2$ and let $g(X) = X_1$ with X_1 centered. What is the correct Sobol-Hoeffding decomposition?

$g(X)$	$=$	g_0	$+$	$g_1(X_1)$	$+$	$g_2(X_2)$	$+$	$g_{1,2}(X_1, X_2)$
X_1	$=$	0	$+$	X_1	$+$	0	$+$	0
X_1	$=$	0	$+$	0	$+$	0	$+$	X_1
X_1	$=$	0	$+$	$\frac{1}{2}X_1$	$+$	0	$+$	$\frac{1}{2}X_1$

Variance decomposition

- The non-overlapping condition implies that *$g_I(X_I)$ is orthogonal to $L^2(X_J)$ such that $J \cap I \subsetneq I$*:

$$\begin{aligned}\mathbb{E}[g_I(X_I)h(X_J)] &= \mathbb{E}[\mathbb{E}[g_I(X_I)h_J(X_J)|X_J]] \\ &= \mathbb{E}[h(X_J)\mathbb{E}[g_I(X_I)|X_{J\cap I}]] = 0\end{aligned}$$

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In particular *the decomposition is orthogonal (ANOVA)*:

$$D := \mathrm{Var}(g(X)) = \sum_{I \subseteq \{1,\ldots,d\}} \mathrm{Var}(g_I(X_I))$$

Multivariate decompositions with commuting projections

S.-H. dec. is an example of multivariate decompositions obtained with *(a class of) commuting projections* $\mathcal{P}_1, \ldots, \mathcal{P}_d$ ([Kuo et al., 2010]), here orthogonals:

$$\mathcal{P}_j(g)(x) = \int g(x) d\mu_j(x_j) = \mathbb{E}[g(X)|X_{-j} = x_{-j}]$$

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The form of the decomposition is simply obtained by expansion:

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The non-overlapping condition is written here $\mathcal{P}_i(g_I) = 0$ for all $i \in I$.

We can see that the map $\Pi_I : g \to g_I$ *is an orthogonal projection*.

An example: separable functions

Consider $g(x) = f_1(x_1) \dots f_d(x_d)$, and denote $m_j = \mathbb{E}(X_j)$. Then:

$$g_I(x_I) = \prod_{i \in I} (f_i(x_i) - m_i) \prod_{j \notin I} m_j$$

Example: $d = 2$

$$\begin{aligned} f_1(X_1) f_2(X_2) = m_1 m_2 \quad &+ \quad m_2(f_1(X_1) - m_1) + m_1(f_2(X_2) - m_2) \\ &+ \quad (f_1(X_1) - m_1)(f_2(X_2) - m_2) \end{aligned}$$

Proofs.

1. $(\mathcal{I} - \mathcal{P}_i)$ acts on f by replacing f_i by $f_i - m_i$
   If $j \notin I$, $\mathcal{P}_j$ acts on f by replacing f_j by m_j
2. Expand the product: $g(x) = ((f_1(x_1) - m_1) + m_1) \dots ((f_d(x_d) - m_d) + m_d)$

Sensitivity indices

Sobol indices

- Partial variances: $D_I = \mathrm{Var}(g_I(X_I))$, and *Sobol indices* $S_I = D_I/D$

$$D = \sum_I D_I, \qquad 1 = \sum_I S_I$$

- $D_i^{\text{tot}} = \sum_{J \supseteq \{i\}} D_J$, $\quad S_i^{\text{tot}} = \frac{D_i^{\text{tot}}}{D}$ *Total index*
- $D_I^{\text{tot}} = \sum_{J \supseteq \{I\}} D_J$, $\quad S_I^{\text{tot}} = \frac{D_I^{\text{tot}}}{D}$ *Total interaction, superset importance*

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Derivative Global Sensitivity Measure (DGSM)

$$\nu_i = \int \left(\frac{\partial g(x)}{\partial x_i} \right)^2 d\mu(x), \qquad \nu_I = \int \left(\frac{\partial^{|I|} g(x)}{\partial x_I} \right)^2 d\mu(x)$$

Usage for screening

Assume that:

- g is continuous on $\Delta = [0,1]^d$
- for all i, the support of μ_i contains $[0,1]$

- **Variable screening**
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- **Variable screening**
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- **Interaction screening**
  If either $D_{i,j}^{tot} = 0$ or $\nu_{i,j} = 0$, then $(x_i, x_j) \mapsto g(x)$ is additive

  Total interactions can be visualized on the *FANOVA graph*, where the edge size is proportionnal to the index value.

Illustration on a toy example

8D g-Sobol function, with uniform inputs on $[0, 1]$:

$$g(x) = \prod_{j=1}^{8} \frac{|4x_j - 2| + a_j}{1 + a_j}$$

with $a = c(0, 1, 4.5, 9, 99, 99, 99, 99)$.

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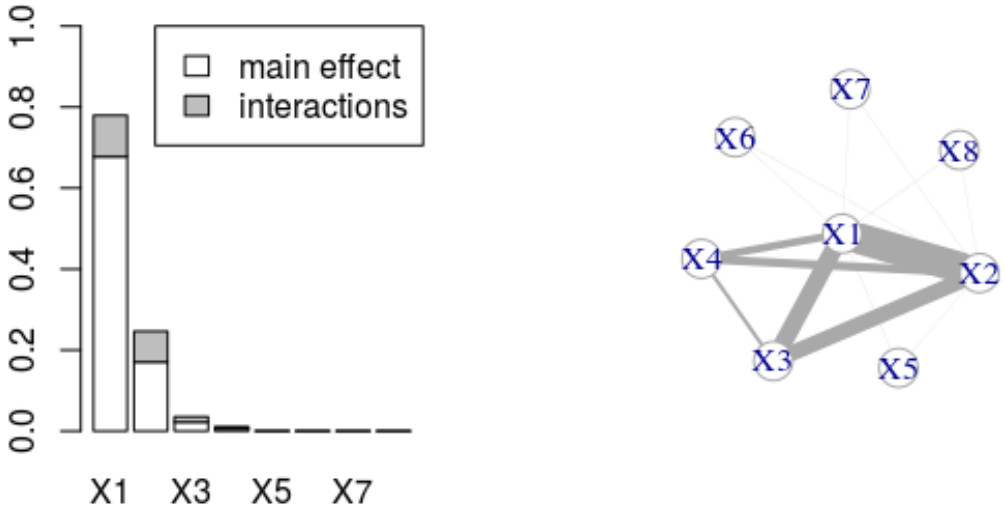

Figure: 1st order analysis (left) and 2nd order analysis (right) with 10^5 simulated data

Illustration on a toy example

A 6D block-additive function, with uniform inputs on $[-1, 1]$:

$$g(x) = \cos([1, x_1, x_2, x_3]^\top \beta) + \sin([1, x_4, x_5, x_6]^\top \gamma))$$

with $\beta = (-0.8, -1.1, 1.1, 1)^\top$ and $\gamma = (-0.5, 0.9, 1, -1.1)^\top$.

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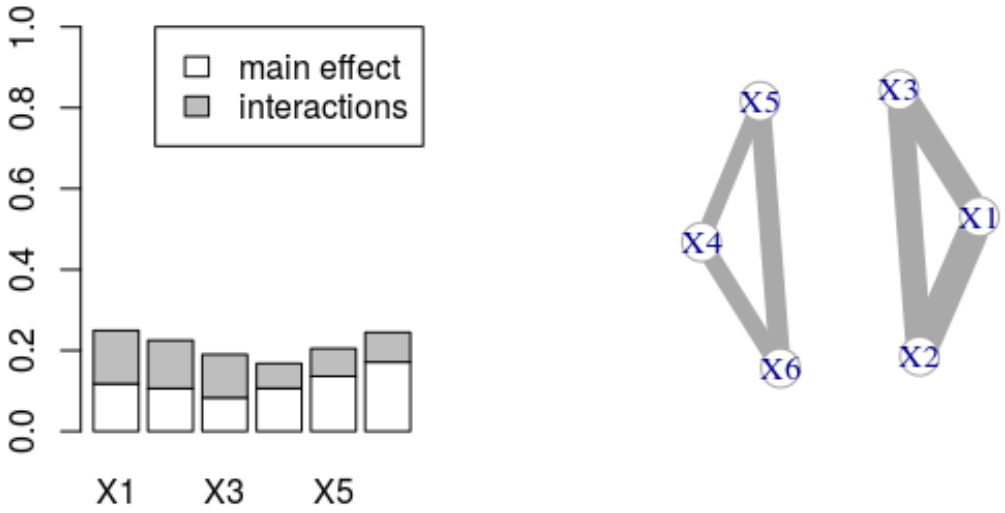

Figure: 1st order analysis (left) and 2nd order analysis (right) with 10^5 simulated data

Advanced topic: estimation of Sobol' indices

Sobol' indices can be computed from the so-called closed Sobol' indices

$$\underline{D}_I(f) = \mathbb{V}\mathrm{ar}(\mathbb{E}[f(X)|X_I]) \qquad \underline{S}_I(f) = \frac{\underline{D}_I}{D}$$

The pick-freeze (PF) estimators *avoid to compute conditional expectations*. They are built on two independent copies X, X' drawn from μ_X, and the evaluations

$$Y = f(X) = f(X_I, X_{-I}), \qquad Y^I = f(X_I, X'_{-I}).$$

Then, it can be proved that:

$$\underline{D}_I(Y) = \mathrm{Cov}(Y, Y^I)$$

Advanced topic: estimation of Sobol' indices

Now, consider two indep. N samples, $X^1, \ldots, X^N$ and $X^{*1}, \ldots, X^{*N}$ of μ. Denote

$$Y^k = f\left(X^k\right), \qquad Y^{I,k} = f\left(X_I^k, X_{-I}^{*k}\right) \qquad (k = 1, \ldots, N).$$

Using the empirical version of mean, covariance leads to the PF estimator

$$\widehat{\underline{S}_I}^{\mathrm{pf}} = \frac{\widehat{\underline{D}_I}^{\mathrm{pf}}}{\widehat{D}^{\mathrm{pf}}}$$

with

$$\begin{aligned}
\widehat{\underline{D}_I}^{\mathrm{pf}} &= \frac{1}{N}\sum_{k=1}^{N} Y^k Y^{I,k} - \left(\widehat{f_0}^{\mathrm{pf}}\right)^2, \quad \widehat{f_0}^{\mathrm{pf}} = \frac{1}{N}\sum_{k=1}^{N}\left[\frac{Y^k + Y^{I,k}}{2}\right] \\
\widehat{D}^{\mathrm{pf}} &= \frac{1}{N}\sum_{k=1}^{N}\left[\frac{(Y^k)^2 + (Y^{I,k})^2}{2}\right] - \left(\widehat{f_0}^{\mathrm{pf}}\right)^2
\end{aligned}$$

This estimator is unbiased, asympt. normal and efficient [Janon et al., 2014, Gamboa et al., 2014]

Part III

References - Thank you for your attention!

Da Veiga, S., Gamboa, F., Iooss, B., and Prieur, C. (2021).
Basics and trends in sensitivity analysis: Theory and practice in R.
SIAM.

Efron, B. and Stein, C. (1981).
The jackknife estimate of variance.
The Annals of Statistics, 9(3):586–596.

Gamboa, F., Janon, A., Klein, T., and Lagnoux, A. (2014).
Sensitivity analysis for multidimensional and functional outputs.
Electronic Journal of Statistics, 8(1):575 – 603.

Hoeffding, W. (1948).
A class of statistics with asymptotically normal distribution.
Ann. Math. Statist., 19(3):293–325.

Iooss, B. (2011).
Revue sur l'analyse de sensibilité globale de modèles numériques.
Journal de la Société Francaise de Statistique, 152:1–23.

Iooss, B. and Lemaitre, P. (2015).
A review on global sensitivity analysis methods.

In Meloni, C. and Dellino, G., editors, *Uncertainty management in Simulation-Optimization of Complex Systems: Algorithms and Applications*, pages 101–122. Springer.

Janon, A., Klein, T., Lagnoux, A., Nodet, M., and Prieur, C. (2014).
Asymptotic normality and efficiency of two Sobol' index estimators.
ESAIM: Probability and Statistics, 18:342–364.

Kuo, F., Sloan, I., Wasilkowski, G., and H.Woźniakowski (2010).
On decompositions of multivariate functions.
Mathematics of computation, 79(270):953–966.

Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., and Tarantola, S. (2008).
Global sensitivity analysis: The primer.
John Wiley & Sons.

Sobol, I. (1993).
Sensitivity estimates for non linear mathematical models.
Mathematical Modelling and Computational Experiments, 1:407–414.